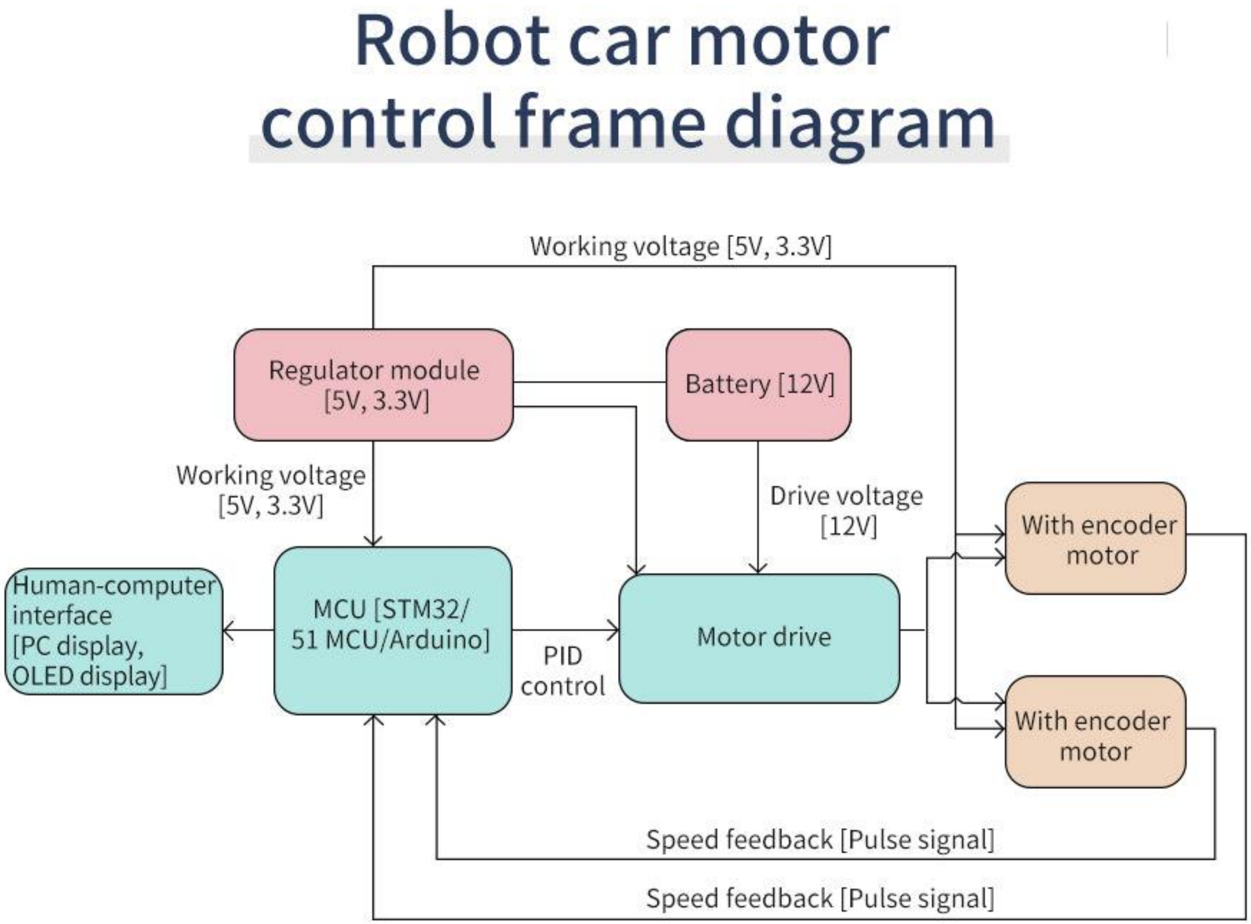
The Robot Motors & Sensors

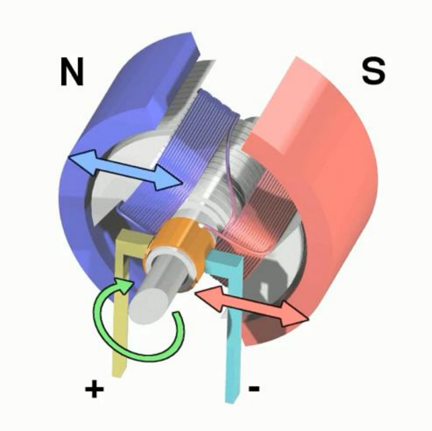
Prof. Muthukumar

Robot Connection Diagram



Motors

- Types of Motors
  - AC motor
  - Brushed DC motor
  - Brushless DC motor
  - Geared DC motor
  - Servo motor
  - Stepper motor
  - DC Linear Actuator



- Motor Drivers
  - H-Bridge - **direction control** and the **speed control**.

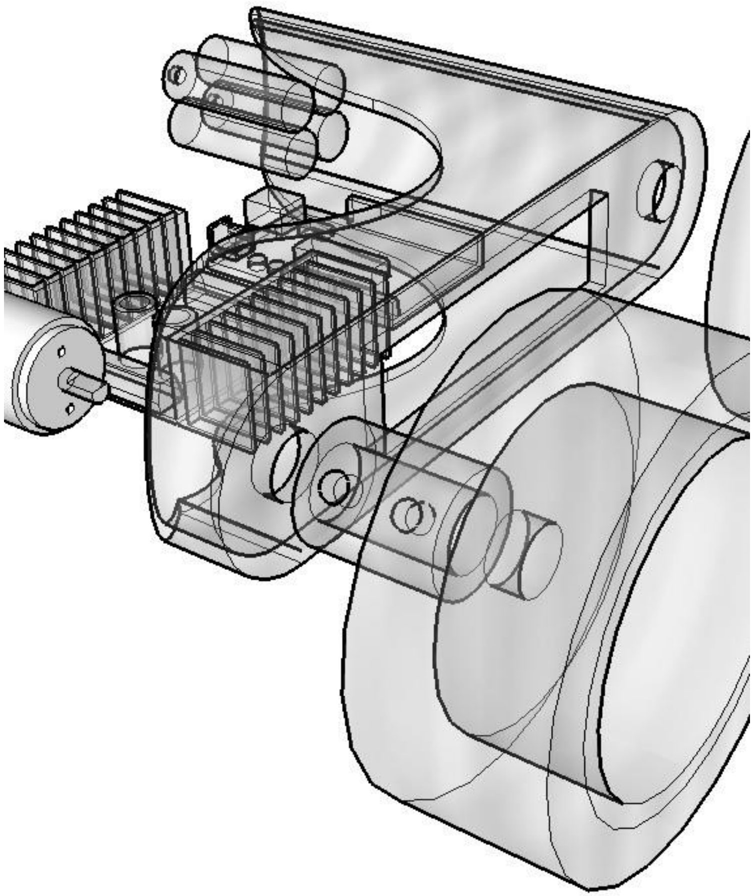
Choose your Motor

- Torque
  - Torque is a measure of a motor's ability to provide a "turning force".
  - rotational force at a distance from the rotational axis.
  - lb-in (pounds inch) in imperial or N-m (newton metre) in metric.
- Speed
- Precision and Accuracy
- Voltage
- Cost
- Form Factor
- Temperature
- Wheel Diameter

- Gears
  - Harmonic Drives

    https://youtu.be/bzRh672peNk
  - Planetary Drives
    - https://youtu.be/gYW8qIS2A6g
  - Cycloidal Drives
    - https://youtu.be/MBWkibie_5I

Choose your Motor



- To calculate the required torque, power, current and battery pack required by a wheeled mobile robot
  - a robot (small or large) does not require much torque to move purely horizontally.
  - motors must produce enough torque to overcome any imperfections in the surface or wheels, as well as friction in the motor itself.

https://community.robotshop.com/tutorials/show/drive-motor-sizing-tool

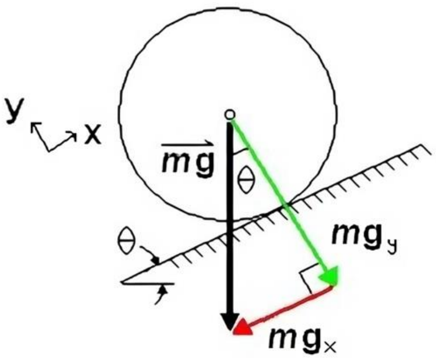
Motor Torque

- On an inclined surface (at an angle theta)
  - one component of its weight (**mg_x** parallel to the surface) causes the robot to move downwards.
  - The other component, **mg_y** is balanced by the normal force the surface exerts on the wheels.

$$mg_x = mg^*\sin(\theta)$$
$$mg_y = mg^*\cos(\theta)$$

Rolling resistance: $c_r \approx 0.02$)rubber on hard surface)

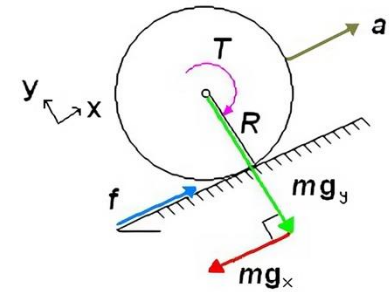
$$mg_y = mg.\ c_r\cos\theta$$



- The torque (**T**) required is:

  T = f * R

- Need torque to accelerate (a) up/forward



Motor Torque

- all forces (**F**) are along the x and y axes. We balance the forces in the x-direction:
- $$\sum F_x = M * a = f - m * g_x$$

$$M * a = \frac{T}{R} - M * g * sin(\theta)$$

$$T = R * M * (a + g * \sin(\theta))$$

- Total Torque T:

$$T = \frac{(a + g * \sin(\theta)) * M * R}{N}$$

- Total number (**N**) of drive wheels to obtain the torque needed for each drive motor.

$$T = \frac{100}{e} * \frac{(a + g * \sin(\theta)) * M * R}{N}$$

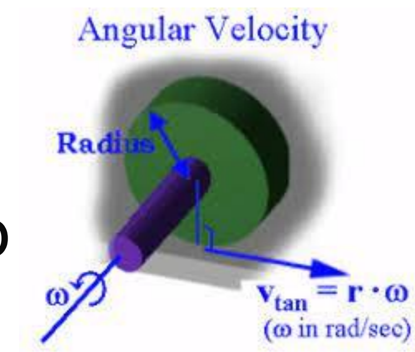
- Considering the efficiency of the Motor (e) can also include *"FactorofSafety"*

Motor Power

- Total power (**P**) per motor can be calculated using the following relation:

$$P = T * \omega$$

- T is known from above and the angular velocity (**ω**). Select the max **ω** to determine max power.



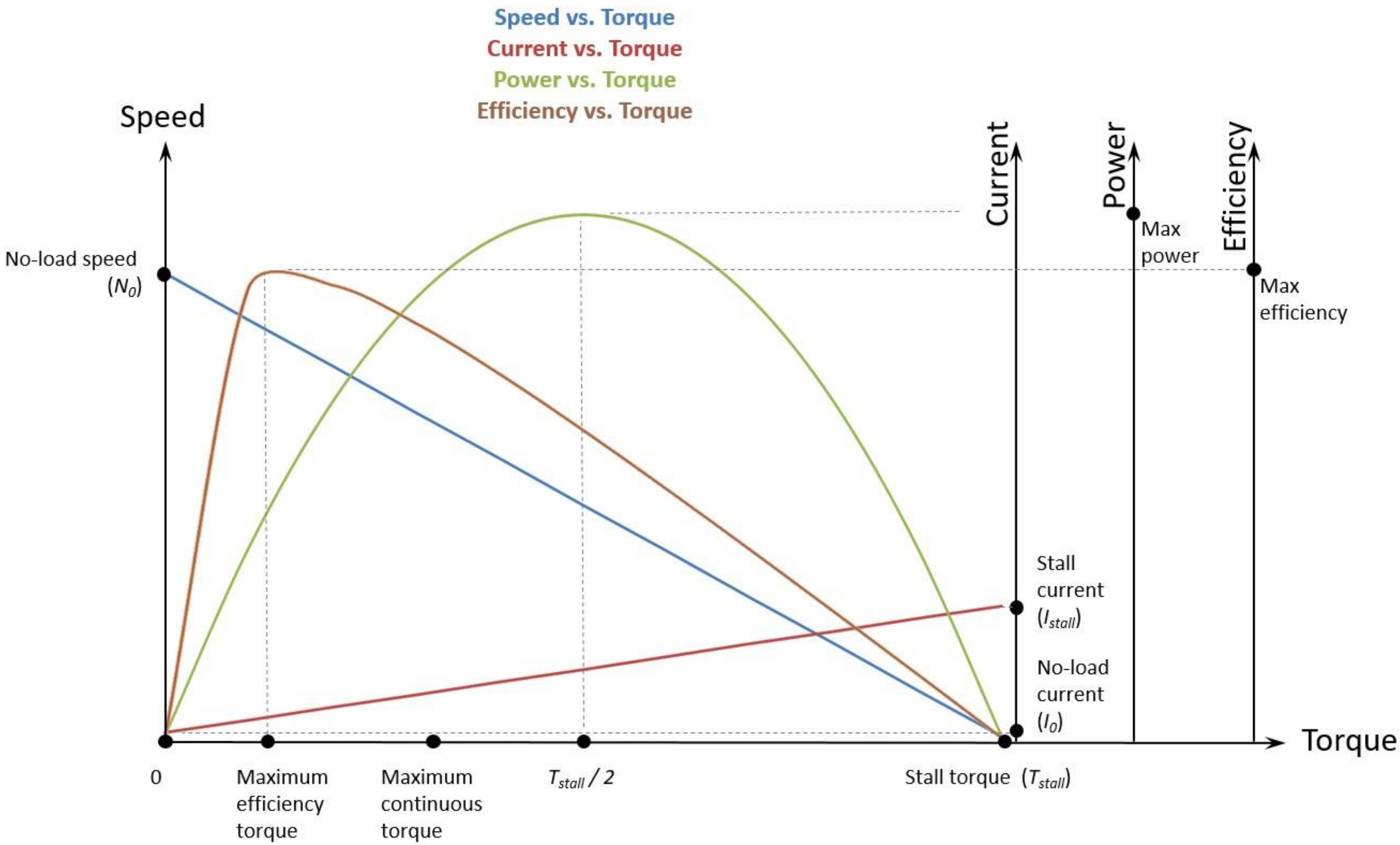
- We can compute max current using the expression

$$P = I * V \qquad I = \frac{T * \omega}{V}$$

- capacity (c) of the battery pack required can be estimated using the equation:

$$c = I * t$$

Motor Characteristics



520 DC Motor

Hall encoder speed measurement

High-precision 11-line magnetic ring

Strong magnetic anti-interference

All metal gears

1:19 **1:30** 1:56

Multiple reduction ratios available

Suitable for a variety of vehicle chassis

Provides application case tutorial

Provide a variety of main control driver codes

Wiring Instructions

1: Motor power cable+
2: Motor power cable -
3: The sensor signal is negative
4: The sensor signal is positive 5V
5: Sensor signal line B phase
6: Sensor signal line A phase

Product parameters

Motor model	MD520Z19_12V	MD520Z30_12V	MD520Z56_12V
Rated motor voltage	12V		
Motor type	Permanent magnet brush		
Output shaft	Diameter 6mm D-type eccentric shaft		
Stall torque	3.1kg·cm	4.8kg·cm	8.3kg·cm
Rated torque	2.2kg·cm	3.3kg·cm	6.5kg·cm
Speed before deceleration	11000rpm	11000rpm	12000rpm
Speed after deceleration	550±10rpm	333±10rpm	205±10rpm
Power rating	≤4W	≤4W	≤4W
Stall current	3A	3A	4A
Rated current	0.3A	0.3A	0.3A
Gear reduction ratio	1:19	1:30	1:56
Encoder type	AB phase incremental Hall encoder		
Encoder supply voltage	3.3-5V		
Number of magnetic loops	11		
Interface type	PH2.0 6Pin		
Function	With its own pull-up shaping, the single-chip microcomputer can directly read the signal pulse		
Single motor weight	150g±1g	150g±1g	150g±1g

Note: The recommended power supply range for a motor with a rated voltage of 12V is between 11V and 16V, and 12V is recommended.

Motor model description

MD 520 Z30_12V

- Rated voltage 12V
- Reduction ratio 1:30
- 520 represents the motor model
- MD stands for Metal DC Motor

- Stall Torque
  - The torque generated by a motor when its output shaft is stationary (zero speed).
- Rated Torque
  - The continuous torque a motor can output at its rated speed and rated voltage without damage.
- Stall Current
  - The amount of current a motor draws when the shaft is prevented from rotating, either by a physical lock or an extremely heavy load.
- Rated Current
  - The maximum current that a motor is designed to draw continuously without overheating or failing.
- Motor Speed = 11000 rpm
- Gear Ratio = 1:30
- Shaft Speed = 11000/30 = 366.67 rpm
- Spec Speed = 330 +/- 10 rpm
- Wheel Speed/Linear Velocity for 65mm wheel diameter:

$$\omega_{\text{motor}} = 11000\,\frac{\text{rev}}{\text{min}} \times \frac{2\pi\ \text{rad}}{1\ \text{rev}} \times \frac{1\ \text{min}}{60\ \text{s}} \approx 1151.92\ \text{rad/s}$$

$$\omega_{\text{wheel}} = \frac{\omega_{\text{motor}}}{30} = \frac{1151.92\ \text{rad/s}}{30} \approx 38.40\ \text{rad/s}$$

$$v = 38.39\ \text{rad/s} \times 0.0325\ \text{m}$$

$$v \approx 1.248\ \text{m/s}$$

Design

- Design a four wheeled mobile robot with required motor torque (kg-cm), power (Watts), and battery capacity (V and mAh) to carry a payload of 10lb at a maximum upward inclination of 30 deg, wheel diameter 65mm, and can travel at a maximum speed of 1.25m/s?

  - **Drive:** 4× driven wheels (one motor per wheel)
  - **Wheel diameter:** 65 mm → radius $r=0.0325$ m
  - **Target top speed:** $v=1.25$ m/s
  - **Slope:** $\theta=30^\circ$
  - **Rolling resistance:** $c_r \approx 0.2$(rubber on hard surface)
  - **Gravity:** $g=9.81$ m/s²
  - **Overall drivetrain+motor efficiency at speed:** $\eta\approx0.7$
  - **Safety margin for losses/terrains:** +20% on required tractive force

Force Calculations

- Wheel RPM at 1.25 m/s on 65 mm wheels:
- RPM $= \frac{v}{2\pi r} \cdot 60 \approx \frac{1.25}{2\pi \cdot 0.0325} \cdot 60 \approx 367\ RPM$
- Required tractive force (steady speed):

$$F = mg(\sin\theta + c_r \cos\theta)$$

- With $m = 10$**kg**, $\theta = 30°$,$c_r = 0.02$:
- Grade: $mg\sin\theta \approx 49.05$N
- Rolling: $mg\ c_r \cos\theta \approx 1.70$N

Base $F \approx 50.75$N → add 20% margin → F_{eff}
≈ 60.9**N**

Motor/Wheel Trouqe Calculations

- Total wheel-ground torque:

$$\tau_{total} = F_{eff} \cdot r \approx 60.9 \times 0.0325 \approx 1.98\ N\text{–}m$$

- Per wheel (4WD):

$$\tau_{wheel} \approx \frac{1.98}{4} \approx 0.495\ N\text{–}m \approx 5.05\ kg\text{–}cm$$

- Recommendation (continuous): ≥ 5 kg·cm per wheel at ~367 RPM at the wheel.
- Recommendation (stall/peak): ≥ 10–12 kg·cm per wheel to cover starts, bumps, and quirks.

Power and Battery

- Power
  - Mechanical power at wheels (steady):

$$P_{mech} = F_{eff} \cdot v \approx 60.9 \times 1.25 \approx 76\text{ W}$$

  - Electrical input (with $\eta \approx 0.7$):

$$P_{elec} \approx \frac{76}{0.7} \approx 109\text{ W}$$

- Per wheel at speed:
  $\tau \cdot \omega = 0.495\ N\text{–}m \times \frac{v}{r} \approx 0.495 \times 38.46 \approx 19\ W.$
- Battery (voltage & capacity)
  - You can run 12 V or 24 V. With this power level, 24 V keeps currents modest and drivers happier.

At speed (avg): $I \approx \frac{109\ W}{24\ V} \approx 4.6\ A$

  - Peaks: design for 10–15 A system bursts
  - Runtime options (at 24 V): [6S LiPo nominal 22.2 V]
  - ~30 min hard climb budget → $\approx 4.6\ A \times 0.5\ h \approx 2.3\ Ah$. With 20% reserve: 24 V, ≈ 3 Ah (≈72Wh).

Power and Battery

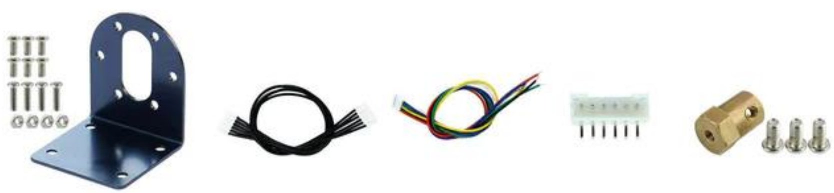
- **4S (14.8 V nominal, 16.8 V full)**

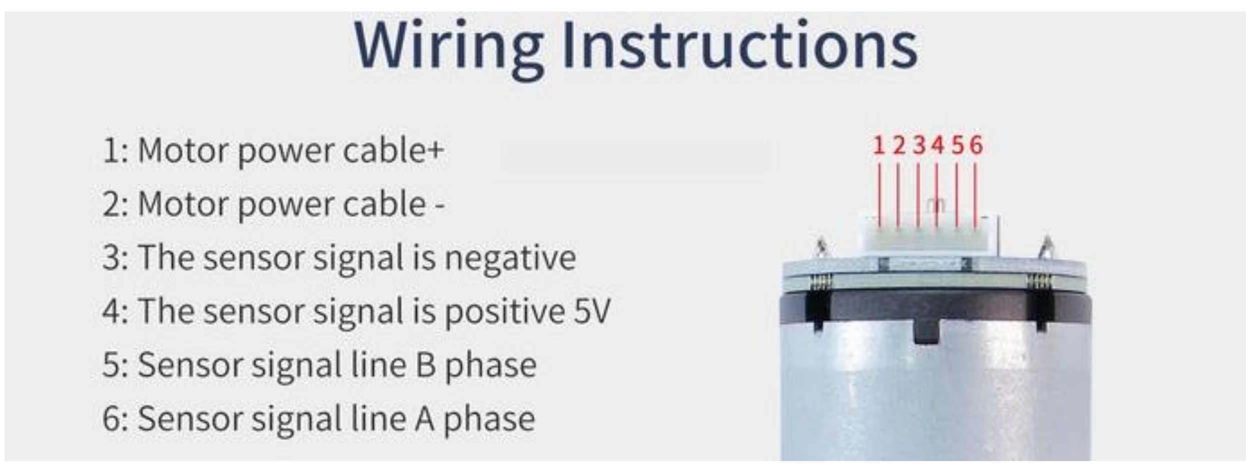
Avg current at speed: $I_{avg} \approx \frac{109\ W}{14.8\ V} \approx 7.36\ A$

Peak current (starts/bumps ~210 W): $I_{peak} \approx \frac{210}{14.8} \approx 14.2\ A$

- **Runtime vs capacity (assume ~80% usable)**
- Runtime $\approx \frac{0.8 \times Ah}{7.36\ A}$.
  - **4S 5000 mAh** (≥20 C) → ~32 min continuous climb; much longer on mixed terrain.

Motor

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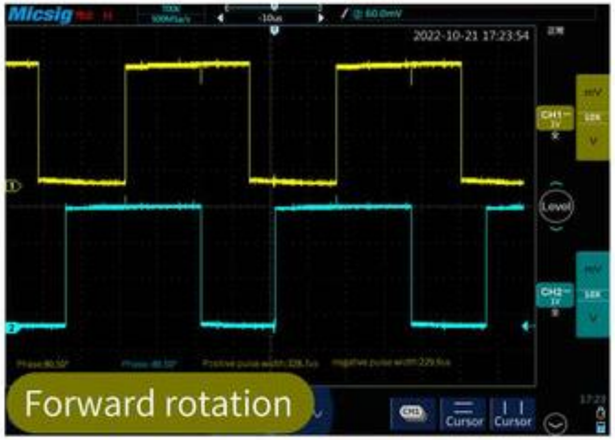
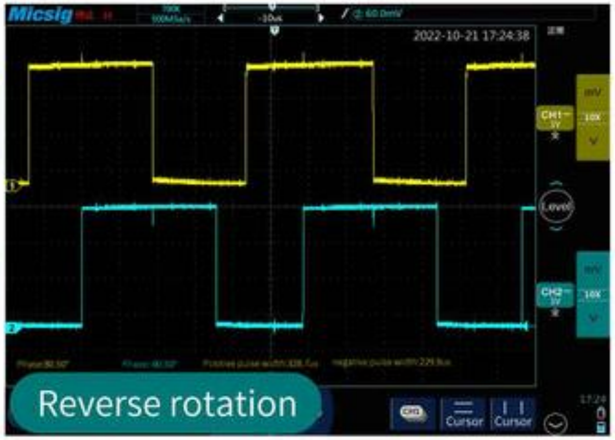
1 2 3 4 5 6

Encoder

The phase difference between the two signals is 100 degrees, and the rotation direction of the motor can be judged according to the sequence of the two signals. The current tire walking distance can be calculated according to the number of signal pulses per unit time and the tire circumference. If only the number of AB-phase pulses per unit time is detected, the speed and slowness of the current motor speed can also be measured.

Take a motor with a reduction ratio of 1:30 as an example, the single-phase output of 11 pulses when the motor rotates one circle, and with a reduction ratio of 1:30, the maximum output of the output shaft of the motor rotates one circle (30*11*4=) 1320 counts. The phase difference of AB two-phase output pulse signal is 100 degrees, which can detect the rotation direction of the motor.





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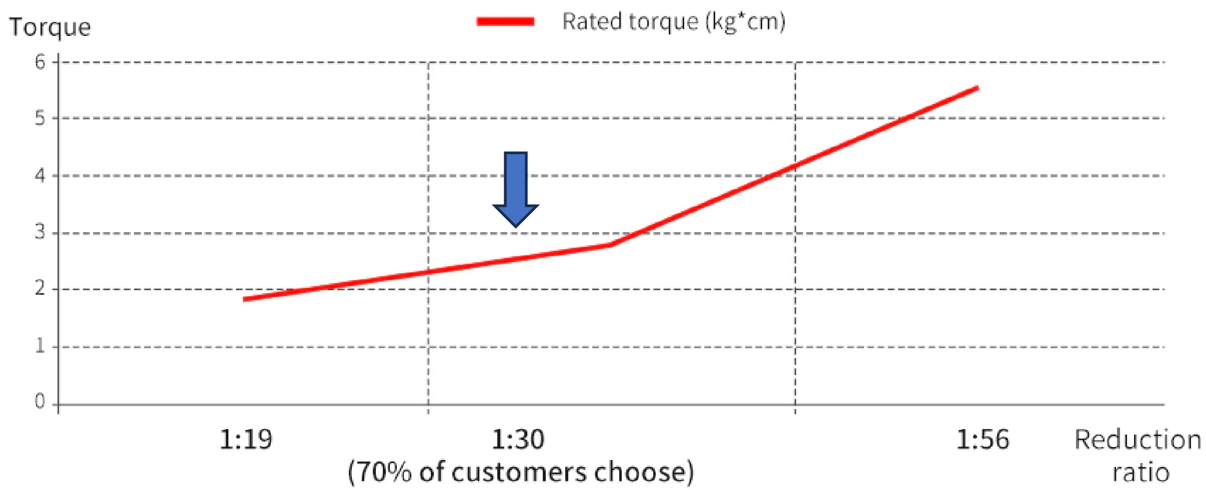
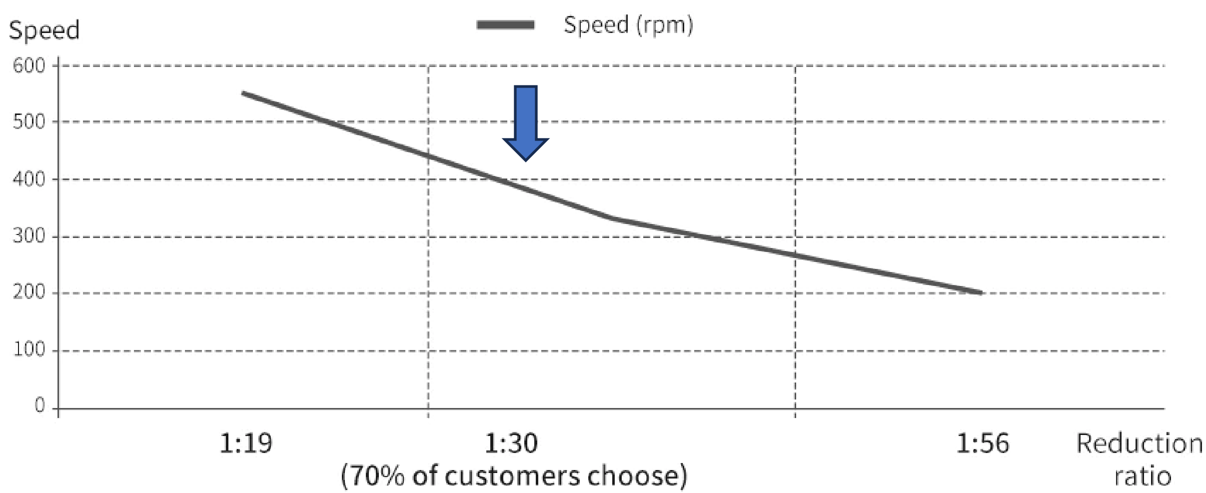
Torque Rating



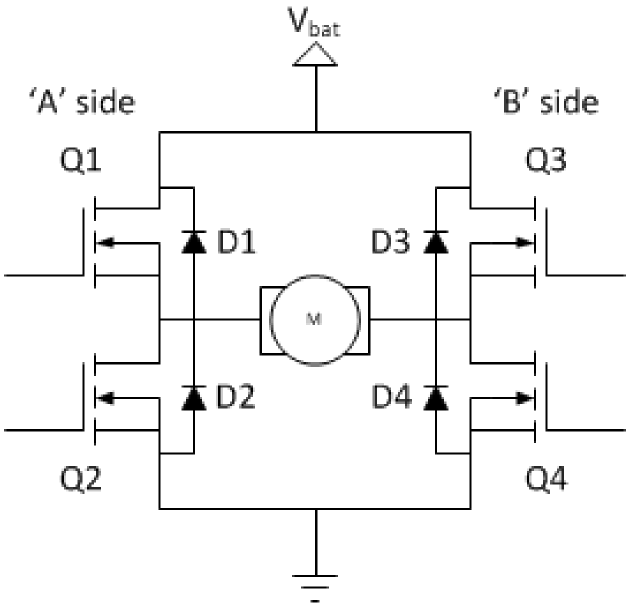


Payload & Speed

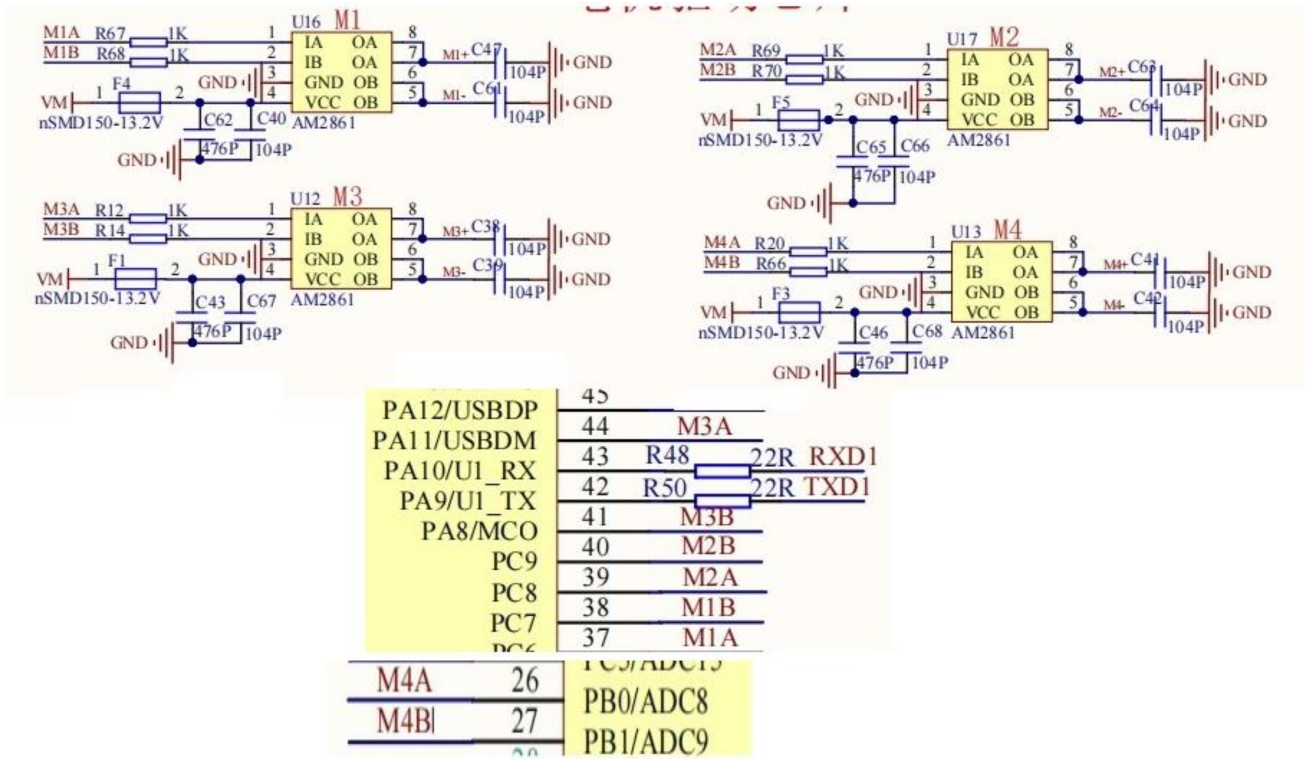
	Ackerman steering car	Differential car	Balance car	Mecanum wheel omnidirectional car	Tracked car
1:19 reduction ratio Max speed	**about 1.8m/s**		about 1.5m/s	1.5m/s	It is not recommended to be the tracked car,the torque is not enough
1:19 reduction ratio Max load capacity	**2-3kg**		0.5-1kg	3-4kg	
1:30 reduction ratio Max speed	**1.2m/s**		**about 1m/s**	**1m/s**	0.3-1m/s
1:30 reduction ratio Max load capacity	**3-4kg**		**2-3kg**	**4-6kg**	5-7kg
1:56 reduction ratio Max speed	0.5m/s [Not recommended]		The speed cannot meet the requirements of the balance car	0.45m/s	**0.2-0.5m/s**
1:56 reduction ratio Max load capacity	4-6kg [Not recommended]			7-10kg	**8-11kg**

Motor Driver

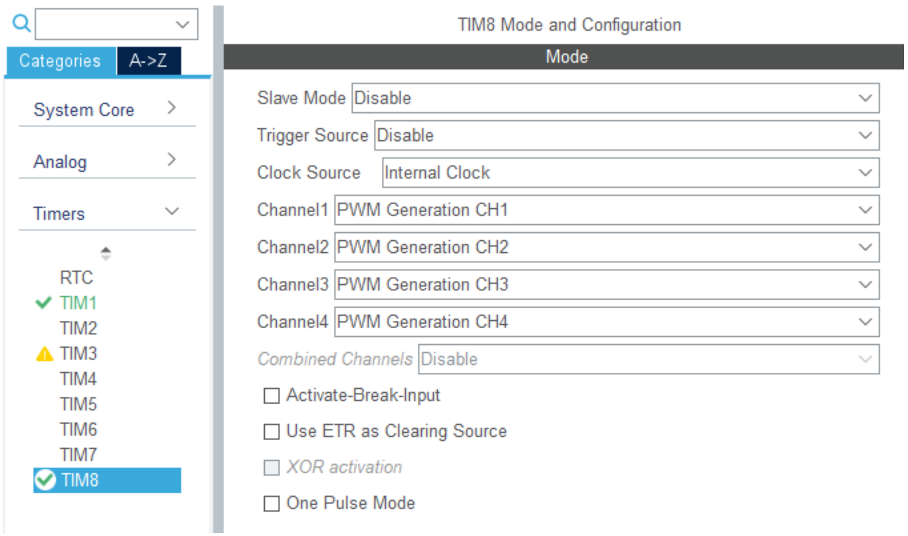
- AM2861
  - **H-Bridge Design:** It uses internal MOSFETs to create an H-bridge for controlling DC motors.
  - **Motor Control:** Allows for controlling the speed (via PWM) and direction of a brushed DC motor.
  - **Current Rating:** Can drive motors with up to 3A continuous current and 5A peak current.
  - **Voltage Range:** Operates from a 4V to 33V voltage range.
  - **Protection:** Features built-in over-temperature and over-current protection.

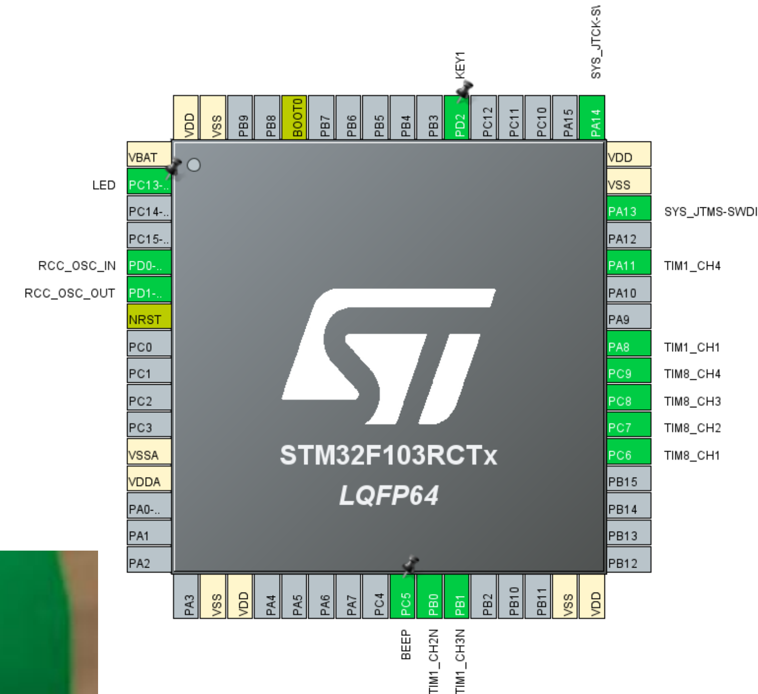


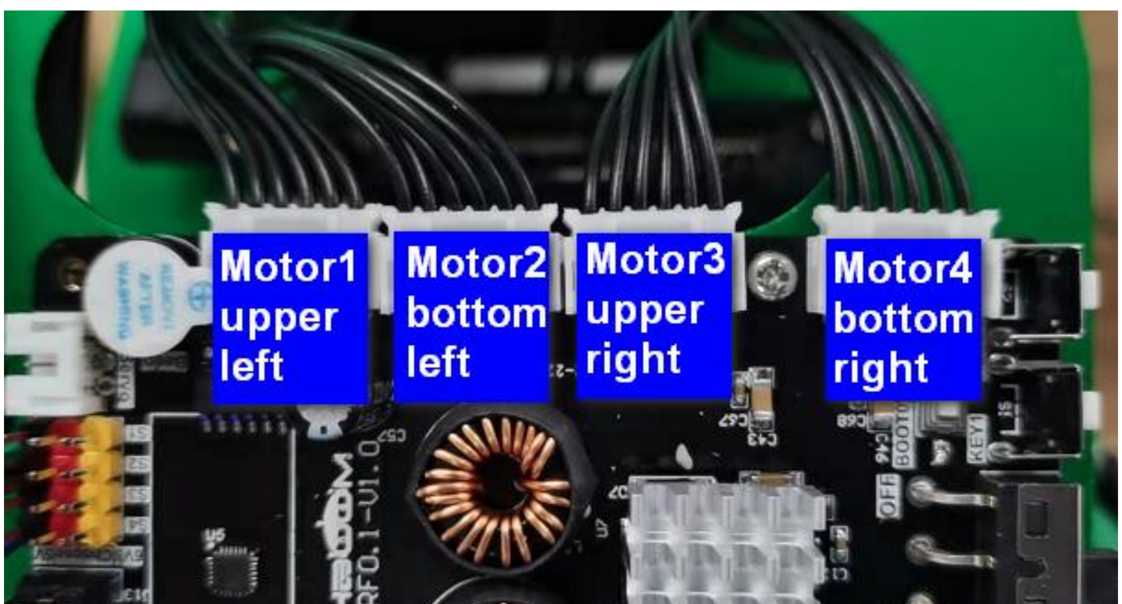
ROS Controller Motor Driver



Motors & ROS controller



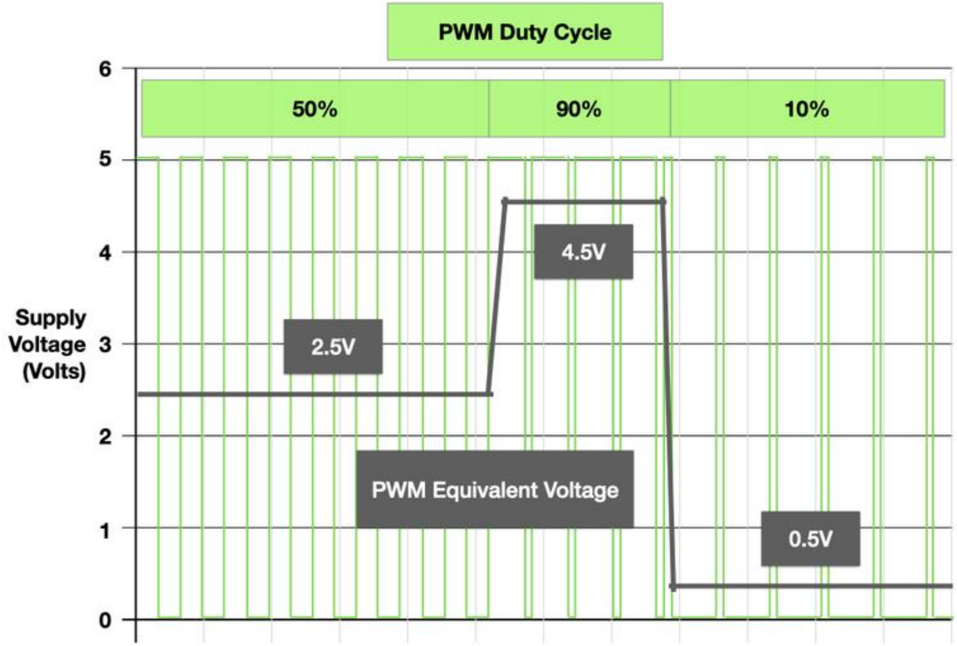




ROS in Motion

Factors in controlling motor speed

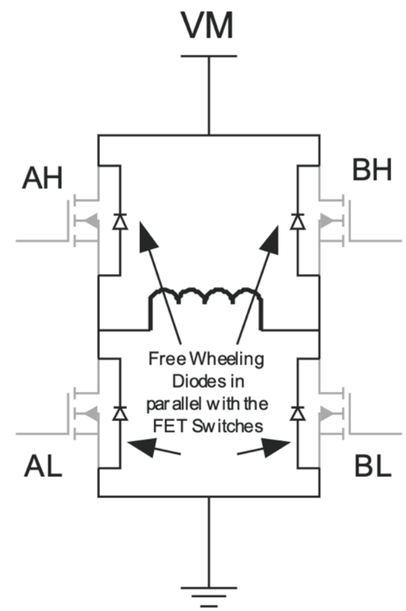
- PWM signal used for a brushed DC motor consists of three primary characteristics, duty cycle, decay mode, and frequency.

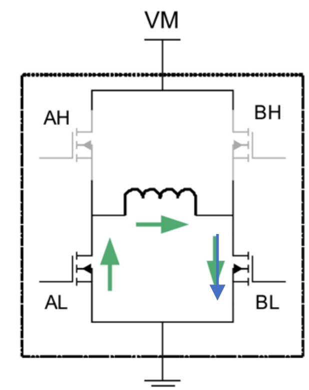


Decay Modes

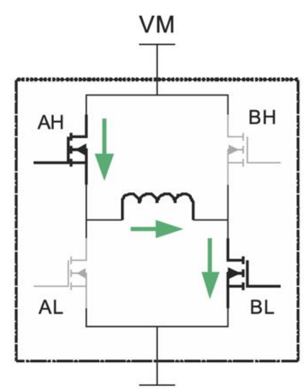
- Fast Decay Mode
  - Or Coasting Mode
- Slow Decay Mode
  - Or Braking Mode

- Async: diodes are used to accept the current flow (back EMFs) while it decays
- Sync: FET ON resistance as a safe path for current decay.

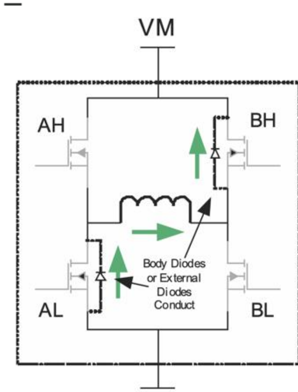




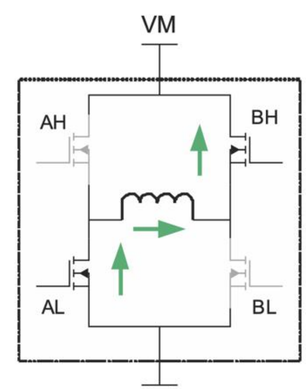
Slow Decay



Normal Operation



During Dead Time



Fast Decay

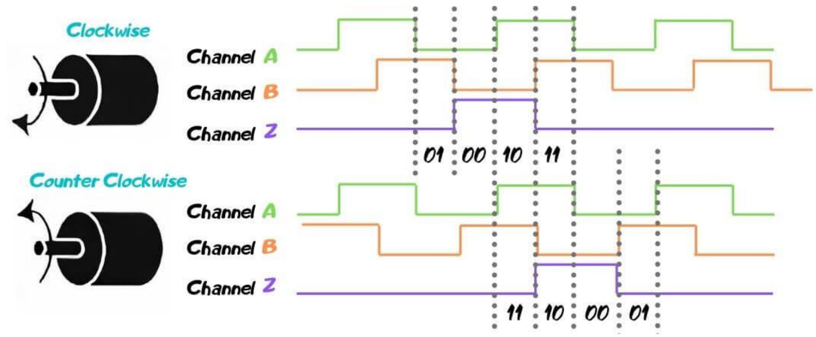
Robotic Sensors

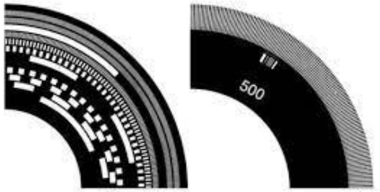
- Wheel Encoders
- Proximity Sensors
  - Ultrasonic
  - Infrared
  - Laser Range Scanner (1D/2D/3D)
- Inertial Measurement Unit (IMU)
  - Gyroscope
  - Accelerometer
  - Magnetometer
- Global Positioning System (GPS)
- Vision Sensor
  - Monocular camera
  - Stereo cameras
- Depth Sensors
  - RGBD Cameras

Encoders

- Speed and Position control

- Types of Encoders
  - Incremental Encoders: The output of an incremental motor encoder is used to control the speed of a motor shaft.
  - Absolute Encoders: The output of an absolute motor encoder indicates both the motion and the position of the motor shaft.



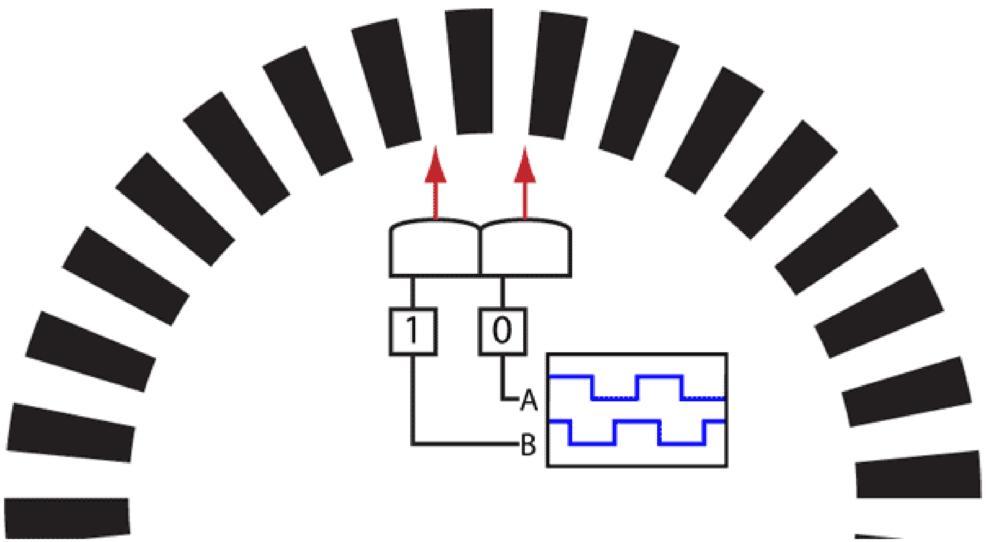


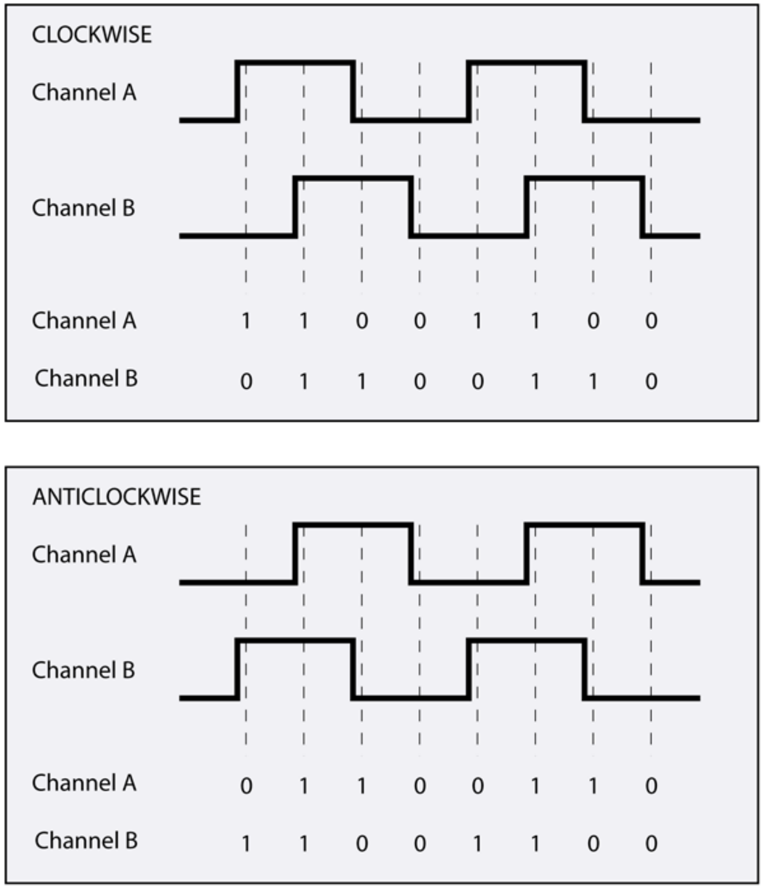
Absolute Encoder

Incremental Encoder

Quadrature Encoders

- A quadrature encoder normally has at least two outputs - Channel A and B



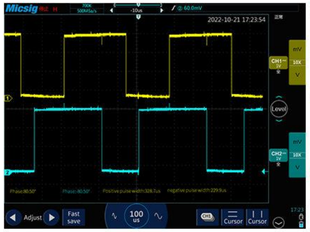


520 DC Motor Encoder

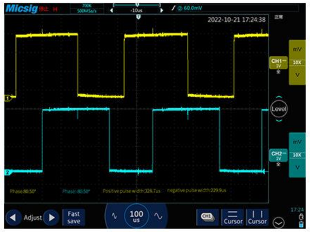
Encoder output description

The phase difference between the two signals is 100 degrees, and the rotation direction of the motor can be judged according to the sequence of the two signals. The current tire walking distance can be calculated according to the number of signal pulses per unit time and the tire circumference. If only the number of AB-phase pulses per unit time is detected, the speed and slowness of the current motor speed can also be measured.

Take a motor with a reduction ratio of 1:30 as an example, the single-phase output of 11 pulses when the motor rotates one circle, and with a reduction ratio of 1:30, the maximum output of the output shaft of the motor rotates one circle (30*11*4=) 1320 counts. The phase difference of AB two-phase output pulse signal is 100 degrees, which can detect the rotation direction of the motor.



Forward rotation



Reverse rotation

Encoder parameters					
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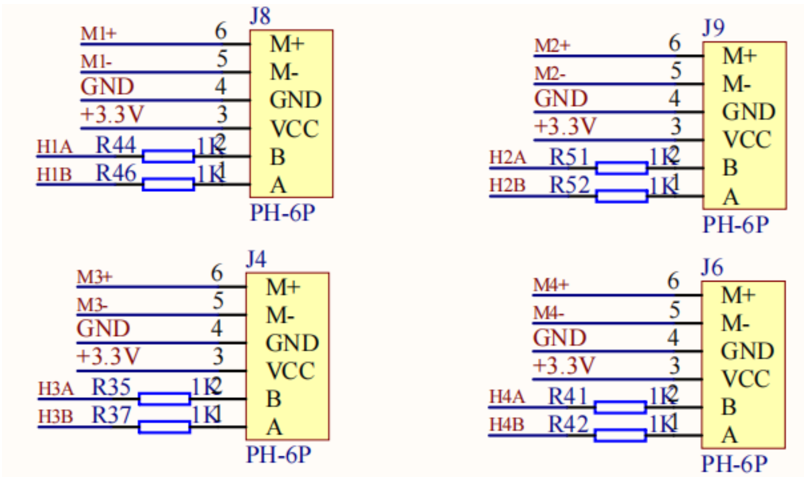
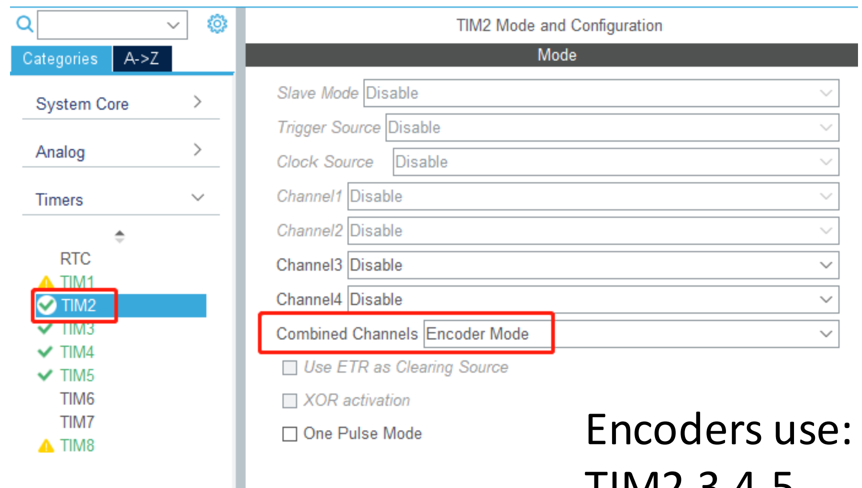
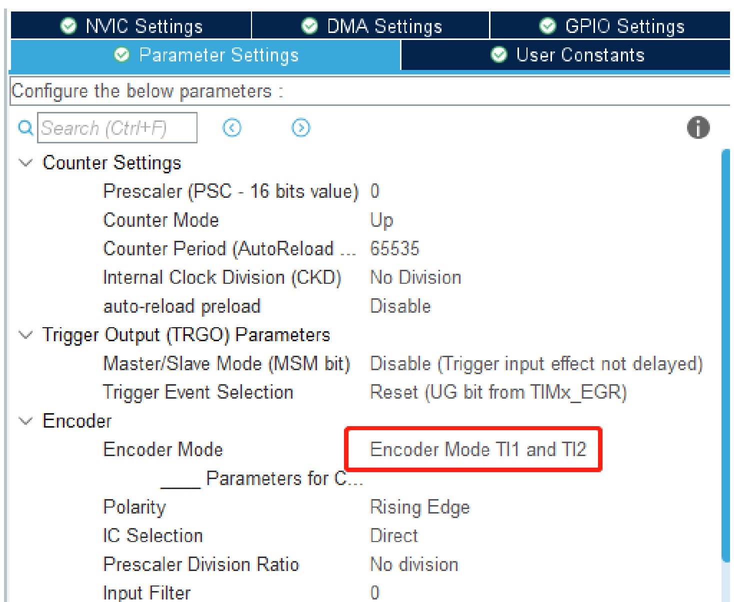
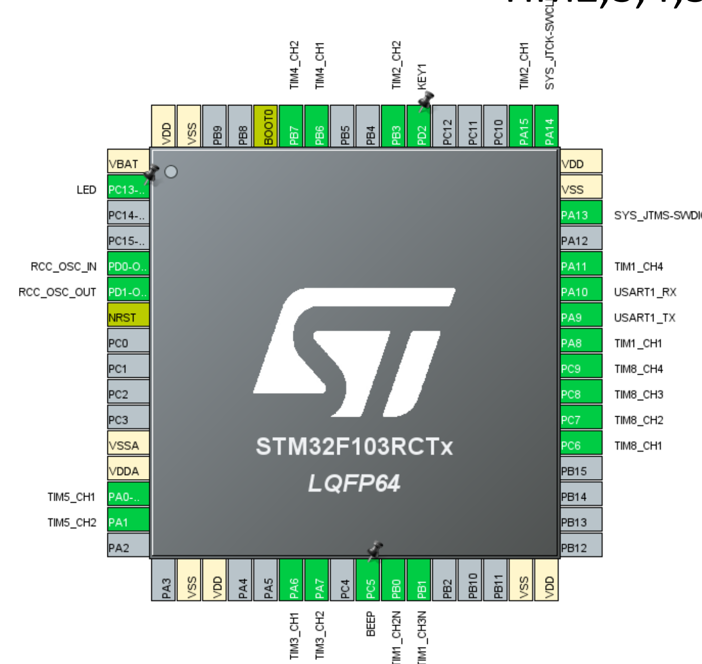
Encoders & ROS Controller





Encoders use:
TIM2,3,4,5

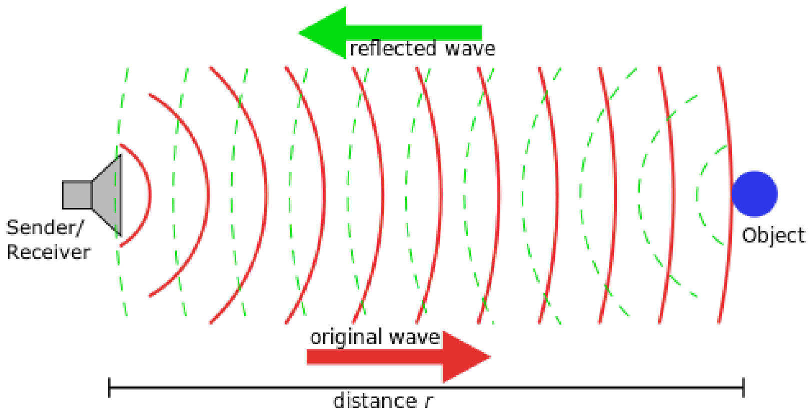




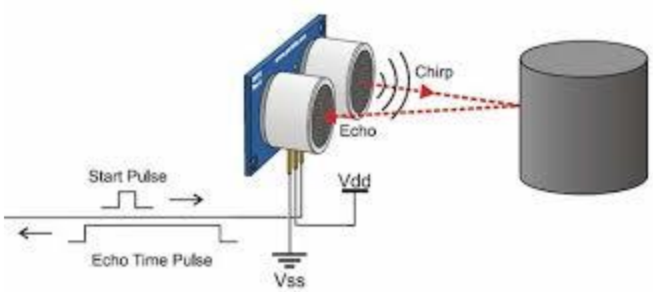
Ultrasound Range Sensors

- **Ultrasonic sensors** measure distances based on transmitting and receiving ultrasonic signals.
- Emit signal to determine distance along a ray
- Make use of propagation speed of ultrasound
- Traveled distance is given by speed of sound (v=340m/s)

$$d = \frac{v\Delta t}{2}$$



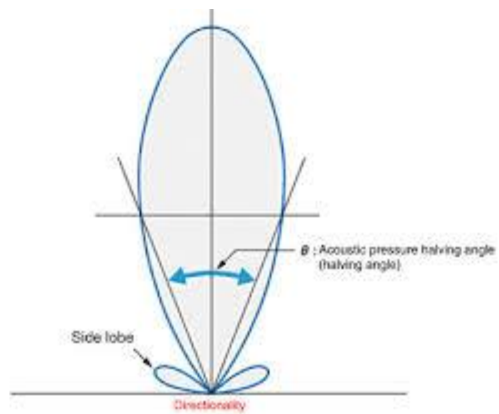
http://en.wikipedia.org/wiki/File:Sonar_Principle_EN.svg

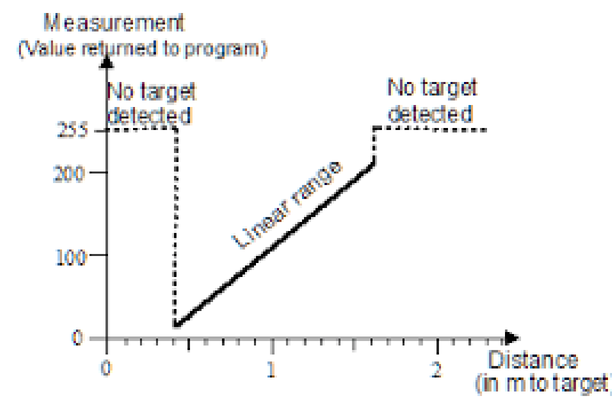


Ultrasound Range Sensors

- Range between 12cm and 5m
- Opening angle around 20 to 40 degrees
- Problems: multi-path propagation, absorption
- Lightweight and cheap

http://www.parallax.com/product/28015

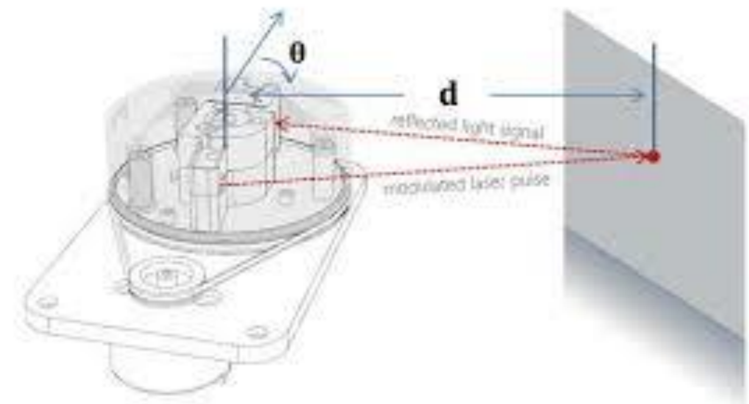




2D LIDAR Scanner

- RPLIDAR A1 is a low cost 360 degree 2D laser scanner (LIDAR). The system can perform 360degree scan within 6meter range. The produced 2D point cloud data can be used in mapping, localization and object/environment modeling.

2D LIDAR Scanner



- For Model A1M8 Only

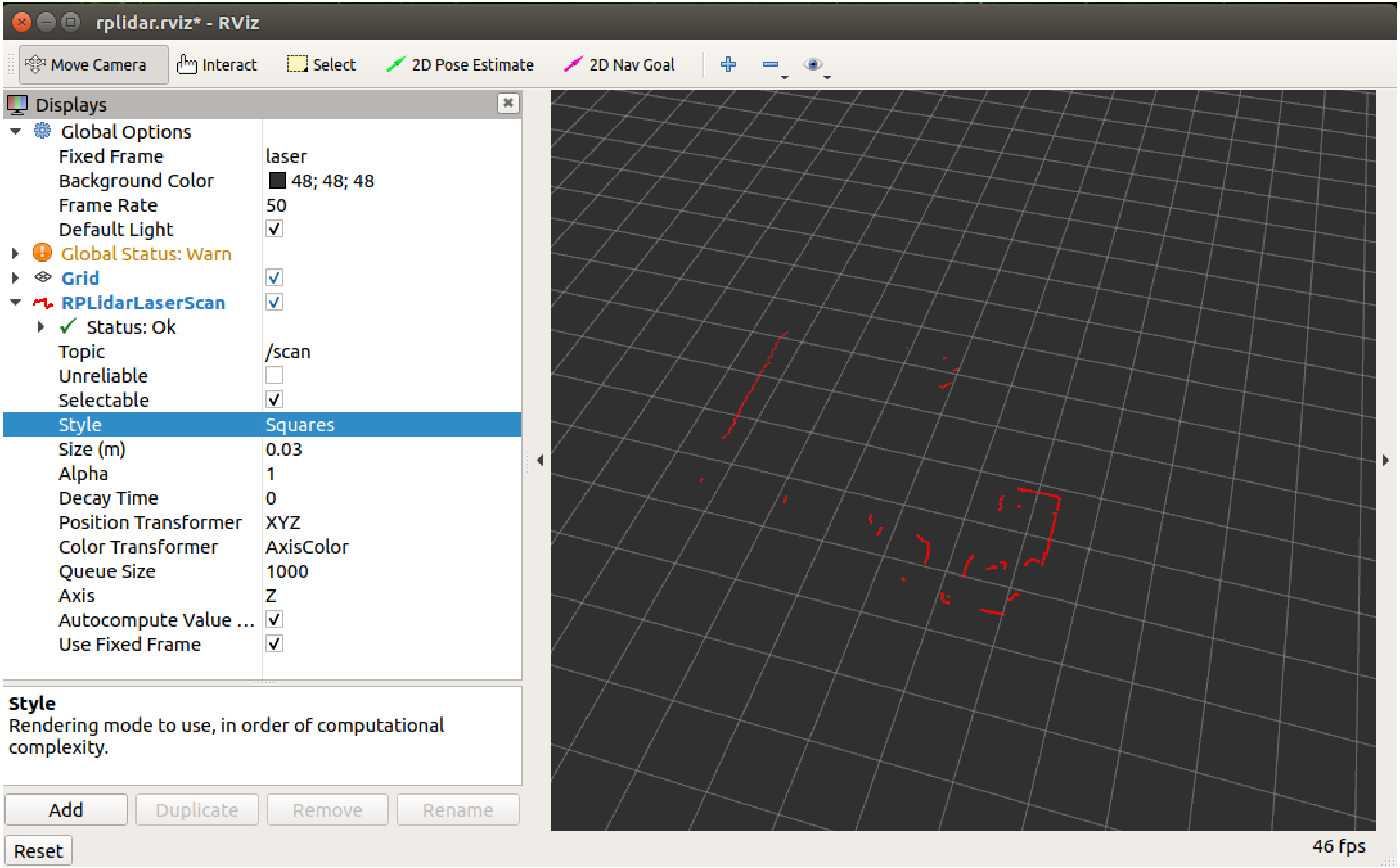
Item	Unit	Min	Typical	Max	Comments
Distance Range	Meter(m)	TBD	0.15-12	TBD	White objects
Angular Range	Degree	n/a	0-360	n/a	
Scan Field Flatness	Degree	-1.5		1.5	
Distance Resolution	mm	n/a	<0.5	n/a	<1.5 meters
			<1% of the distance		All distance range*
Angular Resolution	Degree	n/a	≤1	n/a	5.5Hz scan rate
Sample Duration	Millisecond(ms)	n/a	0.125	n/a	
Sample Frequency	Hz	n/a	≥8000	8010	
Scan Rate	Hz	1	5.5	10	Typical value is measured when RPLIDAR A1 takes 360 samples per scan

Figure 2-1 RPLIDAR A1 Performance

RPLIDAR

- RPLIDAR is a low-cost LIDAR sensor suitable for indoor robotic SLAM application.
- Install the rplidar_ros package
  - https://github.com/Slamtec/rplidar_ros
  - Source the package
- Connect the lidar
  - Remap lidar usb port
    - ./scripts/create_udev_rules.sh
  - Set permission of the lidar device
    - sudo chmod 666 /dev/ttyUSB0
- Start a rplidar node and view the scan result in rviz.
  - $ roslaunch rplidar_ros view_rplidar.launch #for rplidar A1/A2
- Start a rplidar node and run rplidar client process to print the raw scan result
  - $ roslaunch rplidar_ros rplidar.launch #for rplidar A1/A2
  - $ rosrun rplidar_ros rplidarNodeClient

RPLIDAR


rplidar.rviz* - RViz
Move Camera
Interact
Select
2D Pose Estimate
2D Nav Goal
Displays
Global Options
Fixed Frame
laser
Background Color
48; 48; 48
Frame Rate
50
Default Light
Global Status: Warn
Grid
RPLidarLaserScan
Status: Ok
Topic
/scan
Unreliable
Selectable
Style
Squares
Size (m)
0.03
Alpha
1
Decay Time
0
Position Transformer
XYZ
Color Transformer
AxisColor
Queue Size
1000
Axis
Z
Autocompute Value ...
Use Fixed Frame
Style
Rendering mode to use, in order of computational complexity.
Add
Duplicate
Remove
Rename
Reset
46 fps

IMUs

- An inertial measurement unit (IMU) is an electronic device that measures and reports a body's specific force, angular rate, and sometimes the magnetic field surrounding the body, using a combination of **accelerometers** and **gyroscopes**, sometimes also **magnetometers**.
- used to maneuver mobile robots, aircraft, including unmanned aerial vehicles (UAVs), among many others, and spacecraft, including satellites and landers.

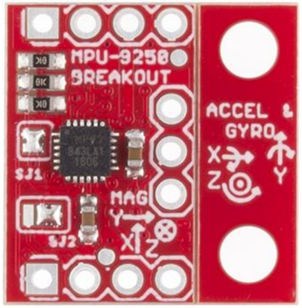
6 DOF Gyro, Accelerometer IMU - MPU6050

- Tri-Axis accelerometer with a programmable full scale range of ±2g, ±4g, ±8g and ±16g
- Combines a 3-axis gyroscope and a 3-axis accelerometer
- I2C Digital-output of 6 or 9-axis Motion Fusion data in rotation matrix
- Working Voltage: 3V - 5V

MPU-9250

- The [MPU-9250](MPU-9250) is the latest 9-axis MEMS sensor from InvenSense®.

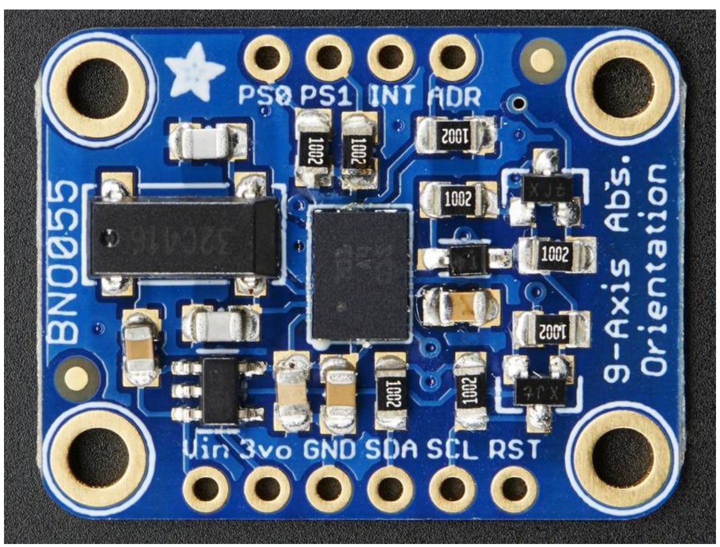
Parameters	MPU-6050	MPU-9250
Accel. full-scale range	±2 g, ±4 g, ±8 g, ±16 g	
Accel. non-linearity	0.5%	0.5%
Accel. cross-axis sensitivity	±2%	±2%
Accel. noise power spectral density	$400\ \mu g/\sqrt{Hz}$	$300\ \mu g/\sqrt{Hz}$
Gyro. full-scale range	±250°/s, ±500°/s, ±1000°/s, ±2000°/s	
Gyro. Non-linearity	0.2%	0.1%
Gyro. cross-axis sensitivity	±2%	±2%
Gyro. rate noise spectral density	$0.005°/s/\sqrt{Hz}$	$0.01°/s/\sqrt{Hz}$
Gyro. total RMS noise	0.05°/s-rms	0.1°/s-rms
Gyro. initial zero tolerance	±20°/s at 25 °C	±30°/s at 25 °C
Magnetometer full-scale range	–	±4800 μT



https://github.com/kriswiner/MPU9250

Adafruit 9-DOF IMU Fusion Breakout - BNO055

- Absolute Orientation (Euler Vector, 100Hz) Three axis orientation data based on a 360° sphere
- Absolute Orientation (Quatenrion, 100Hz) Four point quaternion output for more accurate data manipulation
- Angular Velocity Vector (100Hz) Three axis of 'rotation speed' in rad/s
- Acceleration Vector (100Hz) Three axis of acceleration (gravity + linear motion) in m/s^2
- Magnetic Field Strength Vector (20Hz) Three axis of magnetic field sensing in micro Tesla (uT)
- Linear Acceleration Vector (100Hz) Three axis of linear acceleration data (acceleration minus gravity) in m/s^2
- Gravity Vector (100Hz) Three axis of gravitational acceleration (minus any movement) in m/s^2
- Temperature (1Hz) Ambient temperature in degrees celsius



•**ADR**: Set this pin high to change the default I2C address for the BNO055 if you need to connect two ICs on the same I2C bus. The default address is 0x28. If this pin is connected to 3V, the address will be 0x29

BNO055

Default I2C Address: 0x28

BNO055 Accelerometer:

- Acceleration ranges ±2g/±4g/±8g/±16
- Low-pass filter bandwidths 1kHz~<8Hz
- Operation modes: normal, suspend, low power, standby, deep suspend

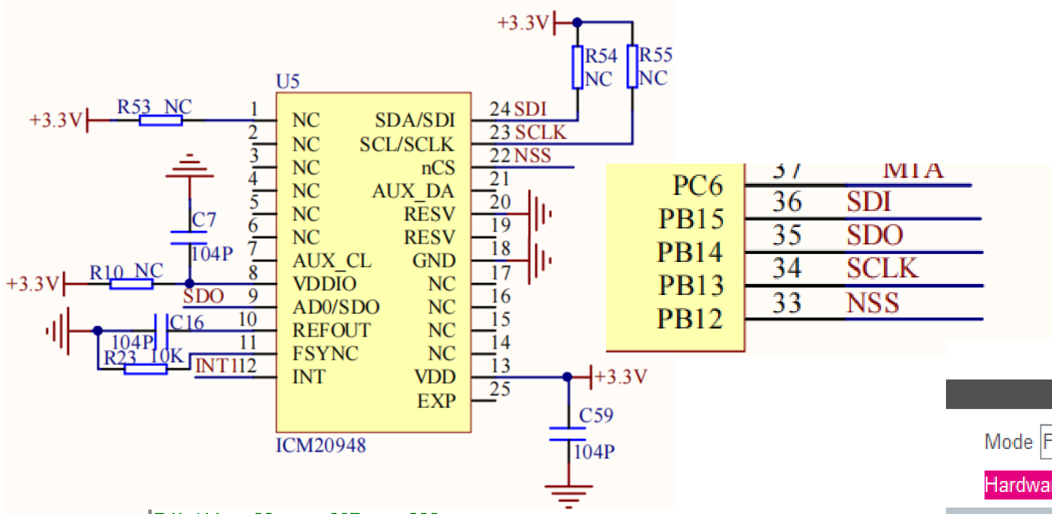
BNO055 Gyroscope:

- Ranges switchable from ±125°/s~2000°/s
- Low-pass filter bandwidths 523Hz~12Hz
- Operation modes: normal, fast power up, deep suspend, suspend, advanced power save.
- On-chip interrupt control: motion-triggered interrupt-signal

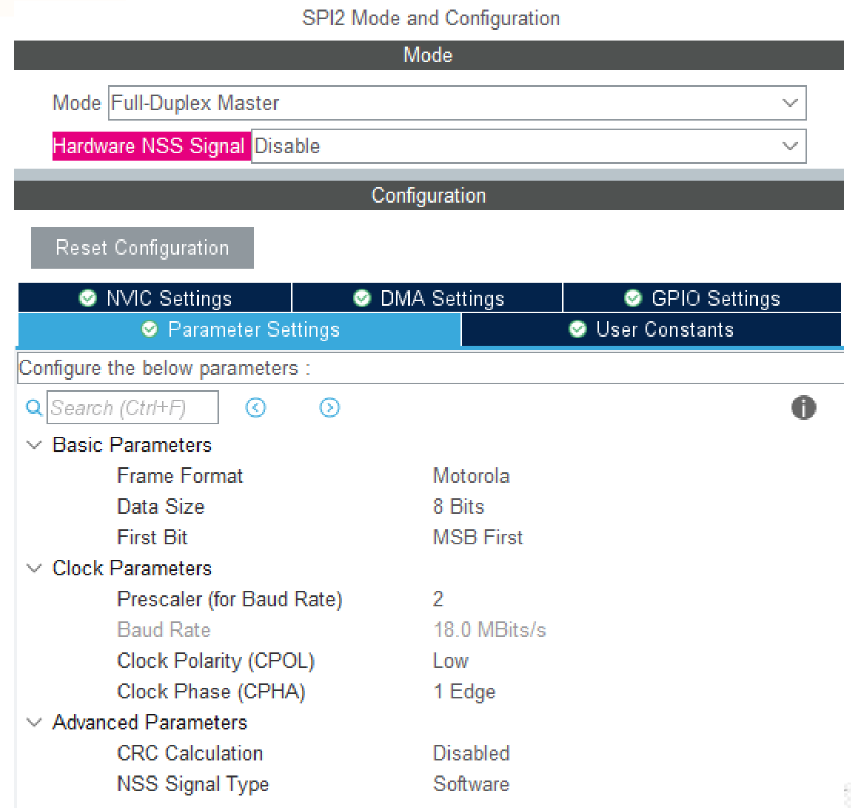
BNO055 Geomagnetic:

- Magnetic field range typical ±1300uT(x-,y-axis);±2500uT(z-axis)
- Magnetic field resolution: ~0.3
- Operating nodes: low power, regular, enhanced regular, high accuracy
- Power modes: normal, sleep, force

ICM20948/IMU & ROS Controller

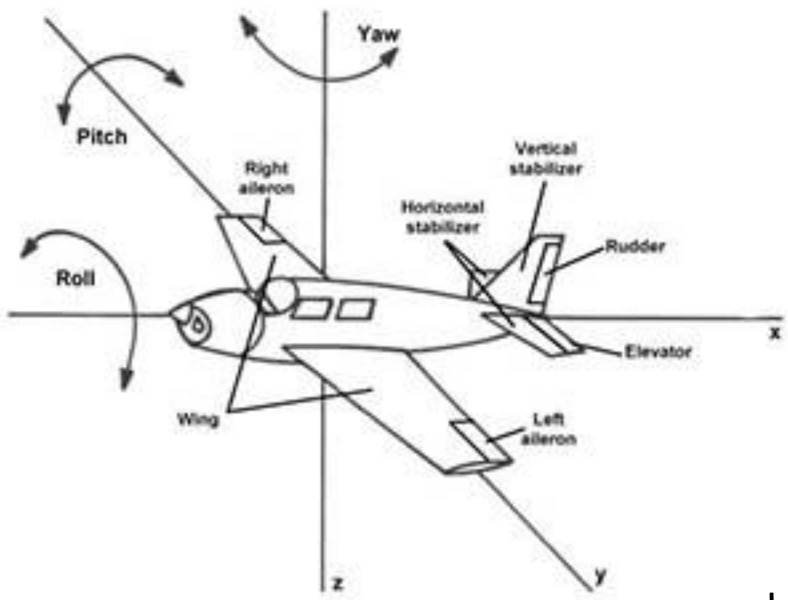


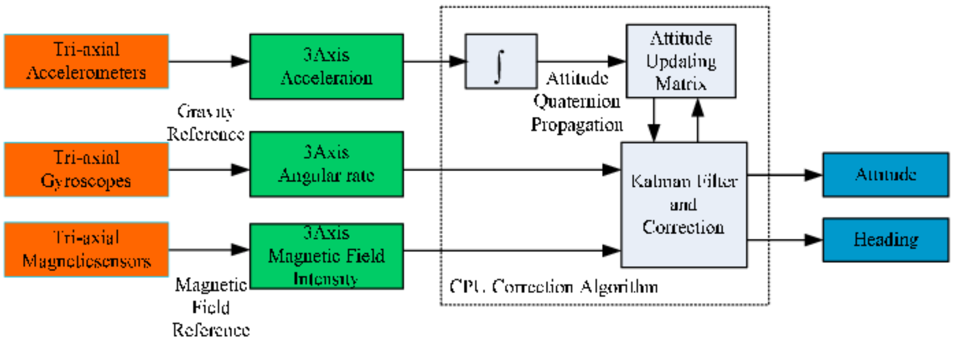

```
RAW Mag:-23,      -265,      608
Axises Accel:0.001953,      0.010254,      1.047363
Axises Gyro:-0.182927,      -0.609756,      -0.060976
Axises Mag:-3.450000,      -39.750000,      91.199997

[2023-06-06 11:04:47.546]# RECV ASCII>
RAW Accel:-6,      6,      2152
RAW Gyro:-3,      -3,      -1
RAW Mag:-11,      -272,      627
Axises Accel:-0.002930,      0.002930,      1.050781
Axises Gyro:-0.182927,      -0.182927,      -0.060976
Axises Mag:-1.650000,      -40.799999,      94.050003

[2023-06-06 11:04:47.655]# RECV ASCII>
RAW Accel:-15,      5,      2158
RAW Gyro:-3,      -7,      -1
RAW Mag:-22,      -269,      609
Axises Accel:-0.007324,      0.002441,      1.053711
Axises Gyro:-0.182927,      -0.426829,      -0.060976
Axises Mag:-3.300000,      -40.349998,      91.349998

[2023-06-06 11:04:47.763]# RECV ASCII>
RAW Accel:-8,      8,      2150
RAW Gyro:-5,      -4,      2
RAW Mag:-28,      -278,      610
Axises Accel:-0.003906,      0.003906,      1.049805
Axises Gyro:-0.304878,      -0.243902,      0.121951
Axises Mag:-4.200000,      -41.700001,      91.500000
```



Attitude and Heading Reference System (AHRS)

- An attitude and heading reference system (AHRS) consists of sensors on three axes that provide attitude information for the robot, including roll, pitch and yaw.





https://ahrs.readthedocs.io/en/latest/

Imu filter madgwick

- **Madgwick Filtering Algorithm** is an advanced orientation estimation technique represented by a quaternion.
- Combines data from multiple sensors (gyro, accel, magnetometer) to reduce noise and drift in real-time.
- corrects the gyroscope drift using accelerometer and magnetometer data.
- Steps:
  - Use gyroscope data to predict orientation (quaternion update).
  - Correct the orientation using accelerometer and magnetometer data.
  - Normalize the quaternion to maintain a valid orientation.
- Parameters:
  - β: Controls the trade-off between responsiveness to the gyroscope and the correction from accelerometer/magnetometer.
  - Sampling Rate Δt: Determines how frequently data is updated.

Summary